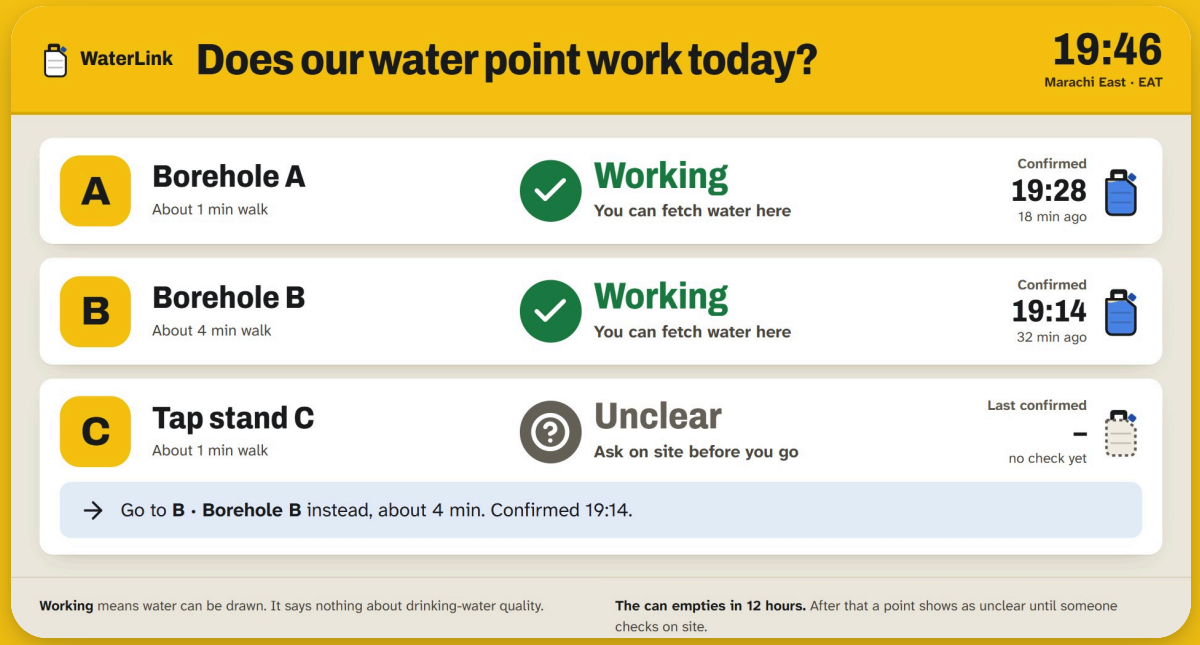
WaterLink

Does our water point work today?

The last confirmed state of every water point before people walk, and a repair that only counts once the pump is tested on site.

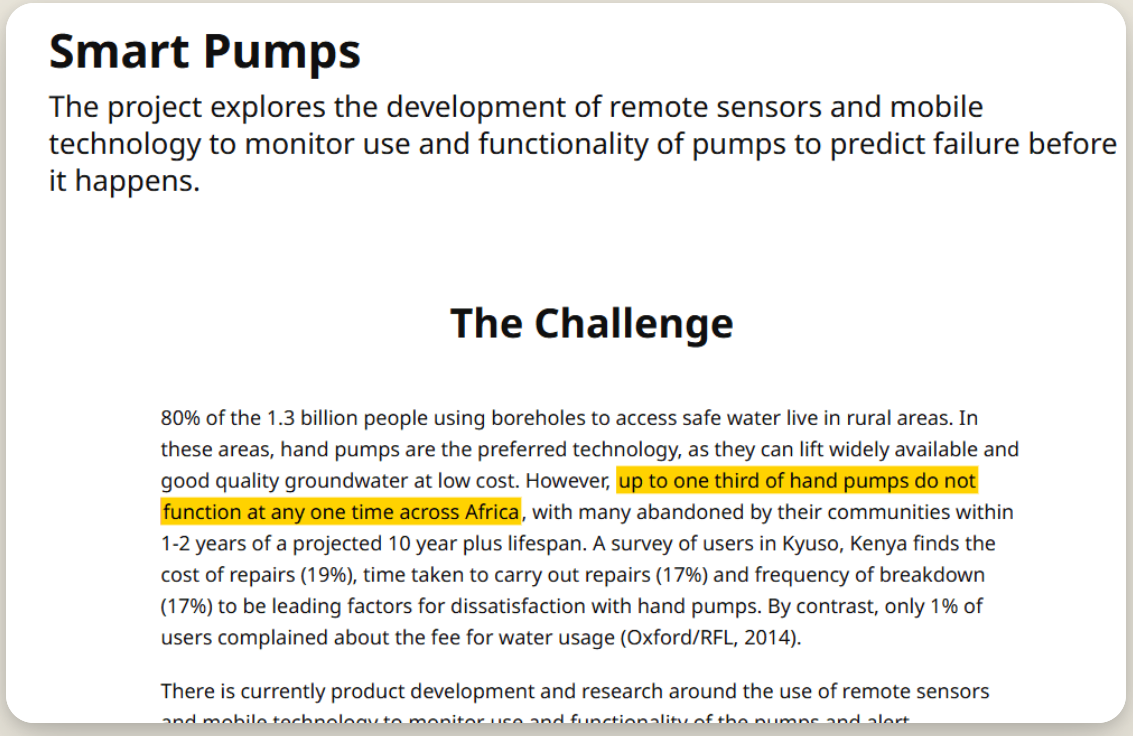


Team 10 · Osman Türkmen, Recep Bas

Zero/One Hackathon · WU Vienna · 22 October 2026 · Track: Social Impact

An active problem, documented by UNICEF

This is not history. UNICEF reports that **up to one in three hand pumps does not work at any one time** across Africa, and the WHO/UNICEF report of 25 August 2025 counts **2.1 billion people** without safely managed drinking water. In Kenya's public data the newest survey is from November 2024, so nobody can see today which pump works.



Smart Pumps

The project explores the development of remote sensors and mobile technology to monitor use and functionality of pumps to predict failure before it happens.

The Challenge

80% of the 1.3 billion people using boreholes to access safe water live in rural areas. In these areas, hand pumps are the preferred technology, as they can lift widely available and good quality groundwater at low cost. However, up to one third of hand pumps do not function at any one time across Africa, with many abandoned by their communities within 1-2 years of a projected 10 year plus lifespan. A survey of users in Kyuso, Kenya finds the cost of repairs (19%), time taken to carry out repairs (17%) and frequency of breakdown (17%) to be leading factors for dissatisfaction with hand pumps. By contrast, only 1% of users complained about the fee for water usage (Oxford/RFL, 2014).

There is currently product development and research around the use of remote sensors

UNICEF Office of Innovation, "Smart Pumps", unicef.org/innovation/smart-pumps. Screenshot of 22 Oct 2026, highlight ours.



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Publications ›

Progress on household drinking water, sanitation and hygiene 2000-2024: Special focus on inequalities

1 in 4 people around the world lack safely managed drinking water

Between 2015 and 2024, 961million people gained access to safely managed drinking water services, increasing global coverage from 68% to 74%. Rural coverage rose from 50% to 60%, while urban coverage remained steady at 83%. Despite this progress, 2.1 billion people still lack safely managed drinking water, 1.4 billion rely on basic services, 287 million on limited services, 302 million use unimproved sources, and

WHO/UNICEF Joint Monitoring Programme, "Progress on household drinking water, sanitation and hygiene 2000–2024", published 25 Aug 2025, data.unicef.org. Screenshot of 22 Oct 2026, highlight ours.

WaterLink

The last confirmed state before the walk, and a repair that counts

A public board.

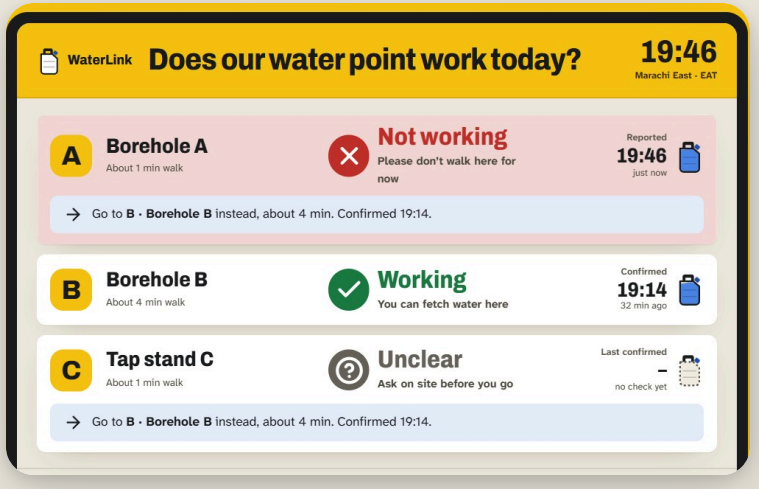
Each water point shows its state, who confirmed it and how long that holds. After 12 hours it reads “unclear”, never an old green.

A repair that counts.

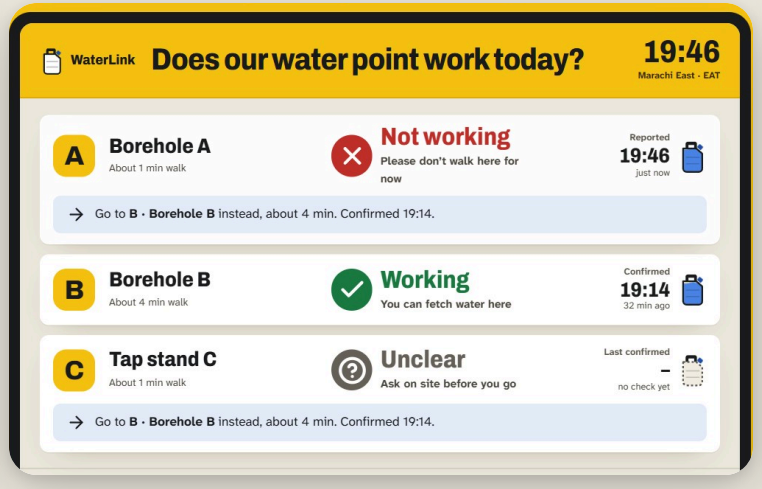
A report becomes a job with a named person and visible blockers. “Repaired” is not “working”: only a test on site turns it green.

A reliable alternative.

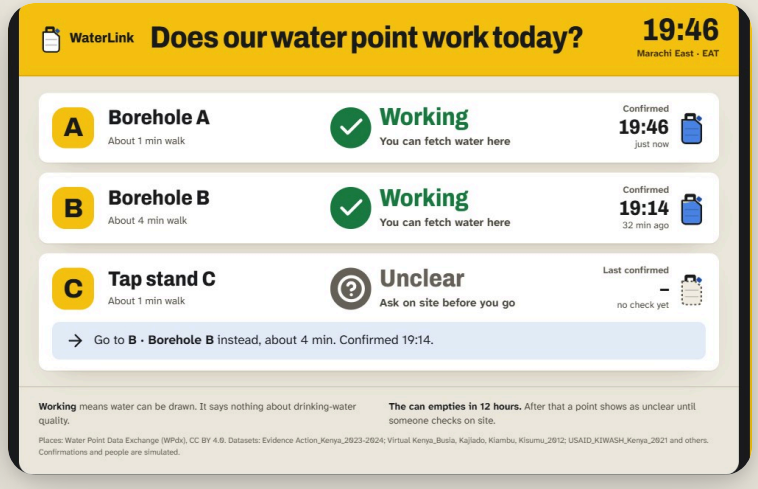
If a point is down, the board names the nearest one, but only if it was freshly confirmed and its route was checked.



Report. A turns red at once and B is named.



Repaired is not working. A stays red until it is tested.

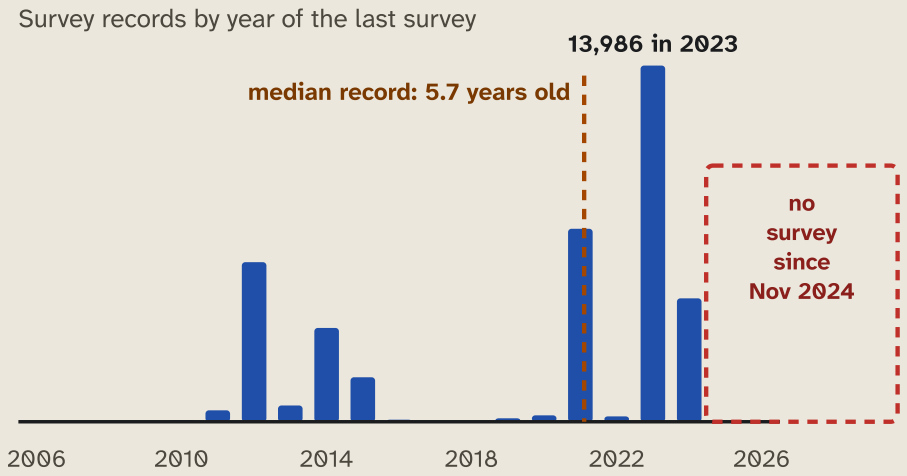


Tested on site. Only now A is green, with the time.

Live and working end to end: board and household view (English, Kiswahili draft), report → job → blockers → repair → test, history, route and access checks, SMS simulation, a replay of real 2014 pump data, practice mode without login. 85 unit and 37 browser tests run on every push. Places come from real surveys (WPdx); people, confirmations and routes are simulated and labelled so.

WaterLink

Numbers behind our assumptions



39,959 survey records for Kenya. The newest is from **30 Nov 2024**, the median is **5.7 years** old, **21%** did not work at the last visit (27% in Busia).

Assumption	Evidence	Status
Families cannot learn today's state	Newest of 39,959 Kenyan records: 30 Nov 2024; median 5.7 years old (WPdx, our calculation).	**Supported**
Many points are broken at any time	Up to 1 in 3 hand pumps in Africa (UNICEF); 21% of Kenyan records non-functional at the last visit.	**Supported**
Waiting for repairs hurts	Kyuso, Kenya: repair cost 19%, repair time 17% and breakdowns 17% top the complaints; 2–3 days without clean water (UNICEF).	**Supported**
An organised repair service works	Kwale and Kitui: downtime from 27 days to under 3 with sensors and a repair service (WASREB 2014).	**Other project**
Households without smartphones can be reached	48.6% of rural and 64.6% of urban people in Kenya own a mobile phone (Housing Survey 2023/24).	**Partly**
Operators confirm often enough; someone pays	No evidence yet. 12 hours is a demo value. No pilot, no impact claim.	**Open**

Sources: Water Point Data Exchange (data.waterpointdata.org, retrieved 22 Oct 2026, our calculation); UNICEF Office of Innovation "Smart Pumps"; WASREB 2014 via REF 2021 impact case study and SDSN TReNDS 2018; Kenya Housing Survey 2023/24 (CA, KNBS).

What we learned, and what comes next

1. **An old green is worse than no answer.**
   Every confirmation expires.
2. **Repaired is not working.**
   Only a test on site turns a point green.
3. **“Working” is not “safe”.**
   No water-quality claim, said on every board.
4. **A dot on the map is not a recommendation.**
   An alternative needs a checked route.
5. **Records are not places.**
   60 survey records turned out to be 42 places.
6. **Say what is unchecked.**
   The Kiswahili is a labelled draft; no impact is claimed.

Next

One neighbourhood, one partner (a county water office, NGO or maintenance service) and four numbers: fresh confirmations, time to a repair that counts, wasted walks, and whether anyone reads “Working” as “clean water”. Baseline first, targets afterwards.

IdeaLAB: we would potentially agree to take part if we are selected.

Try it, no login waterlink-demo.waterlink.workers.dev/?view=stage&sandbox=1

Code github.com/OsmanTuerkmen/WaterLink (a copy is attached to our e-mail)

Team 10 Osman Türkmen, Recep Bas · basreci@gmail.com